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(54) **SUBSTRATE PROCESSING APPARATUS AND METHOD FOR PROCESSING SUBSTRATES**

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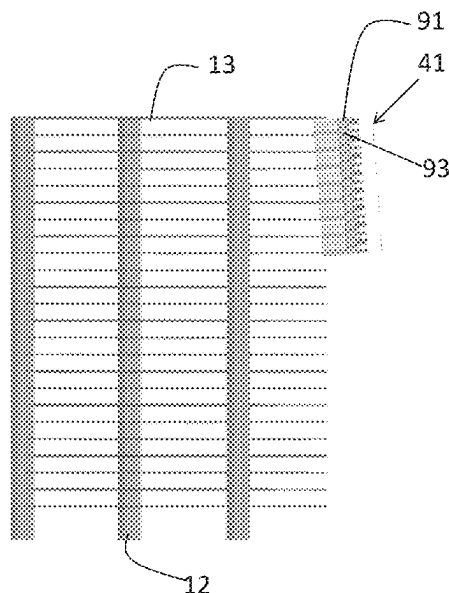
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(57) **ABSTRACT**

The disclosure relates to a substrate processing apparatus, comprising: a first reactor constructed and arranged to process a rack with a plurality of substrates therein; a second reactor constructed and arranged to process a substrate; and, a substrate transfer device constructed and arranged to transfer substrates to and from the first and second reactor. The second reactor may be provided with an illumination system constructed and arranged to irradiate ultraviolet radiation within a range from 100 to 500 nanometers onto a top surface of at least a substrate in the second reactor.

20 Claims, 7 Drawing Sheets



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fig - 1

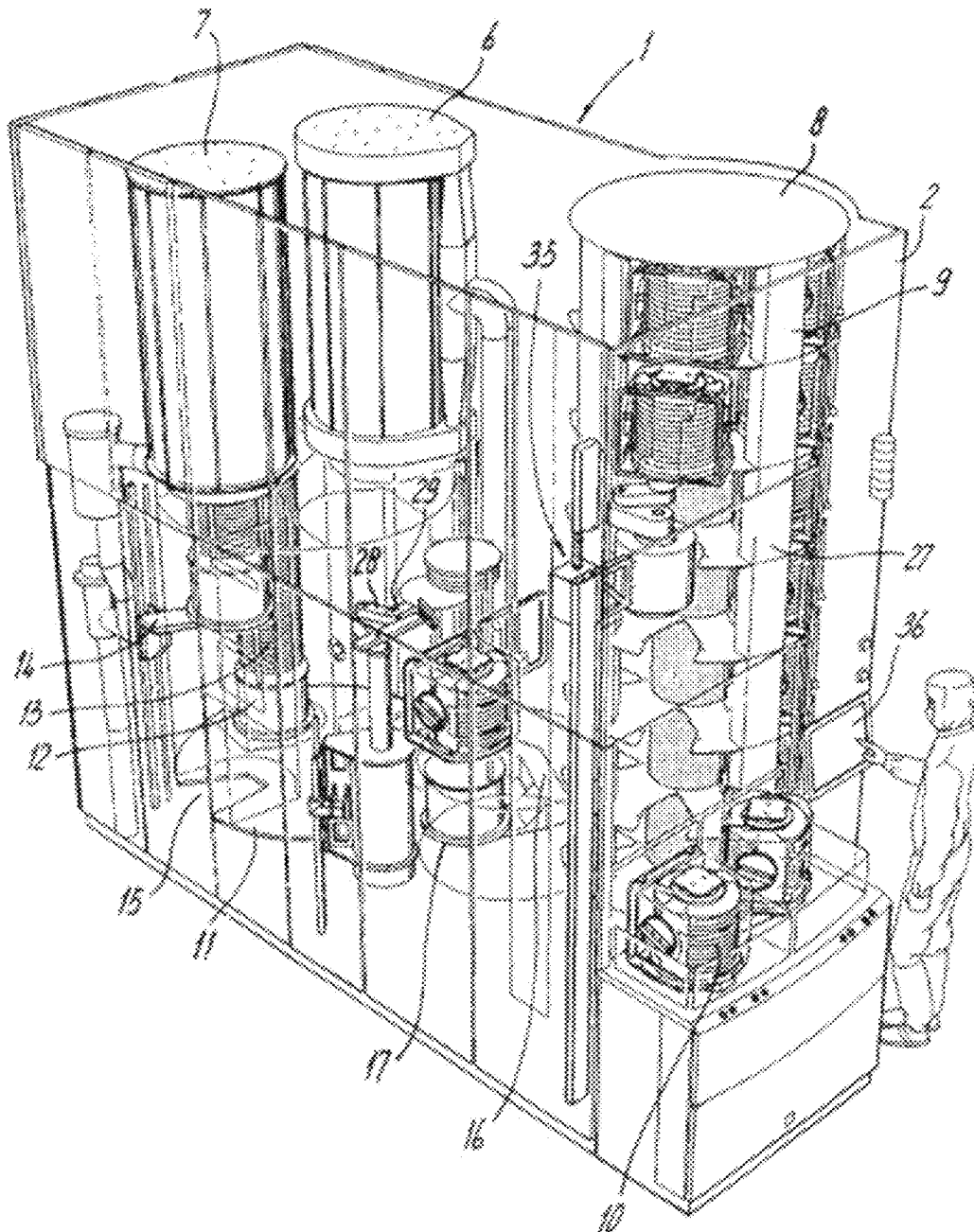
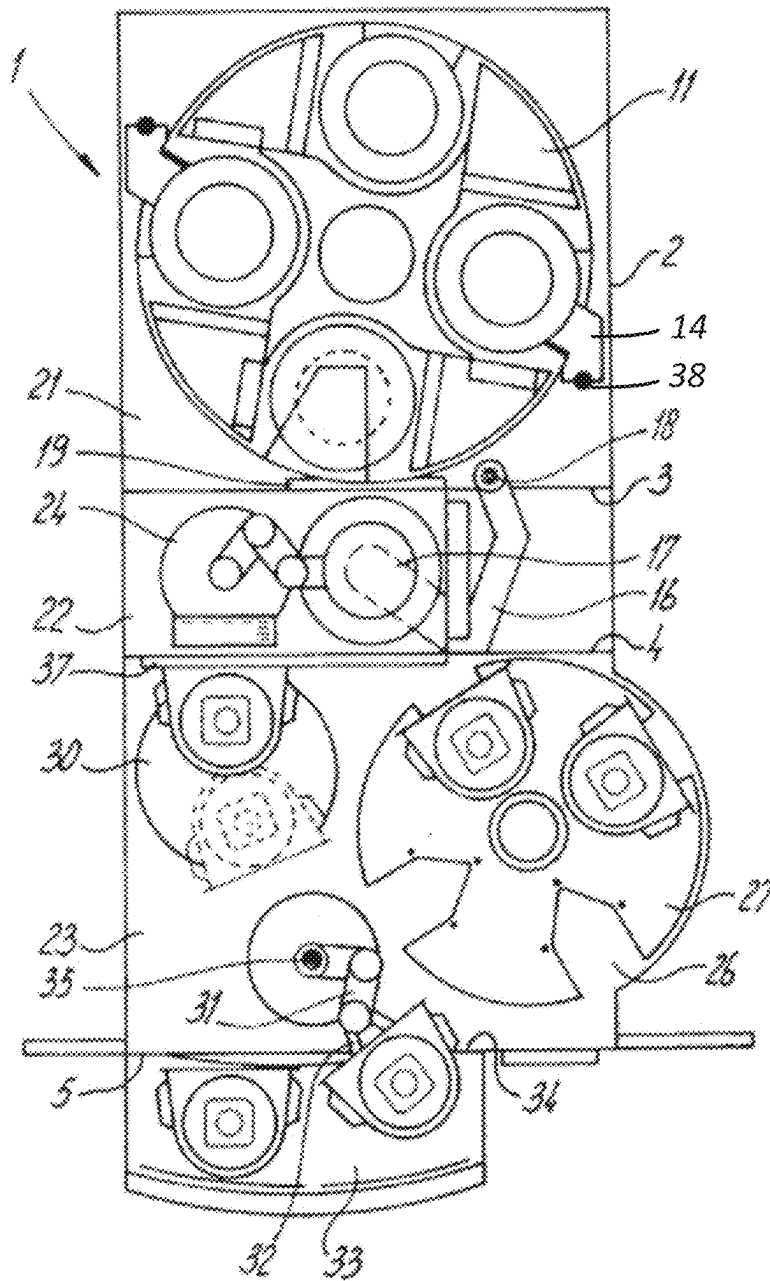


fig - 2



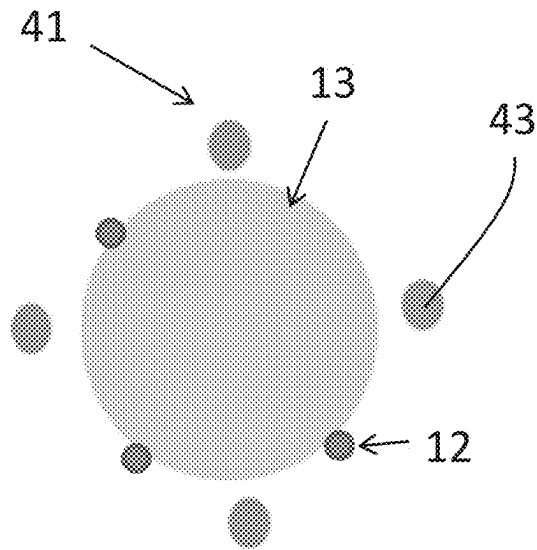


Fig. - 3

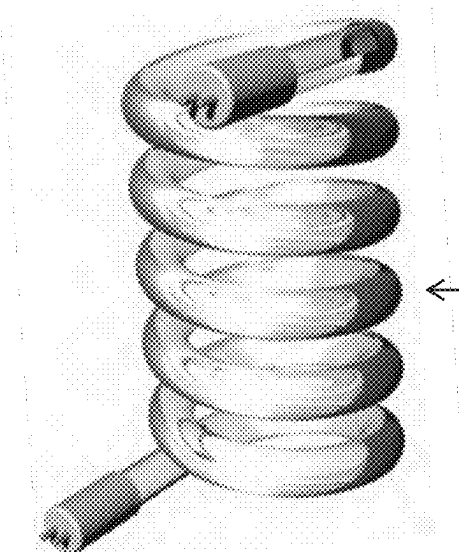


Fig. - 4a

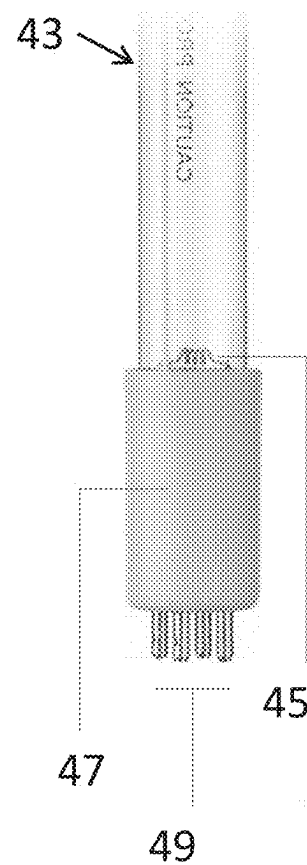
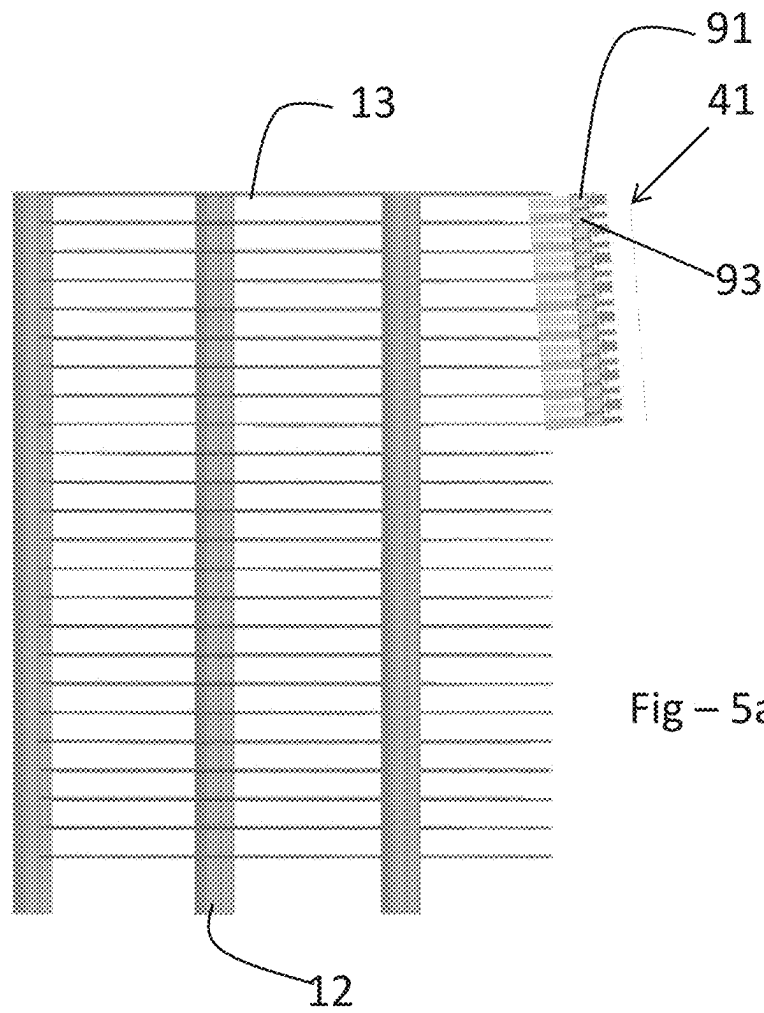
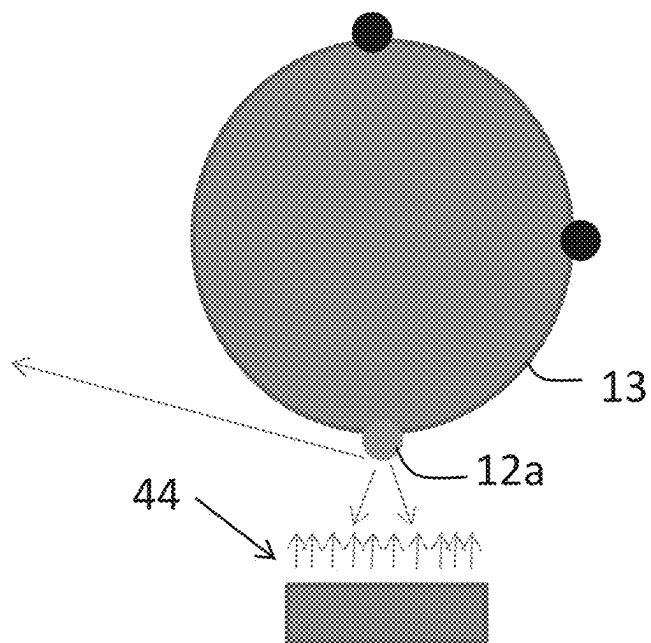
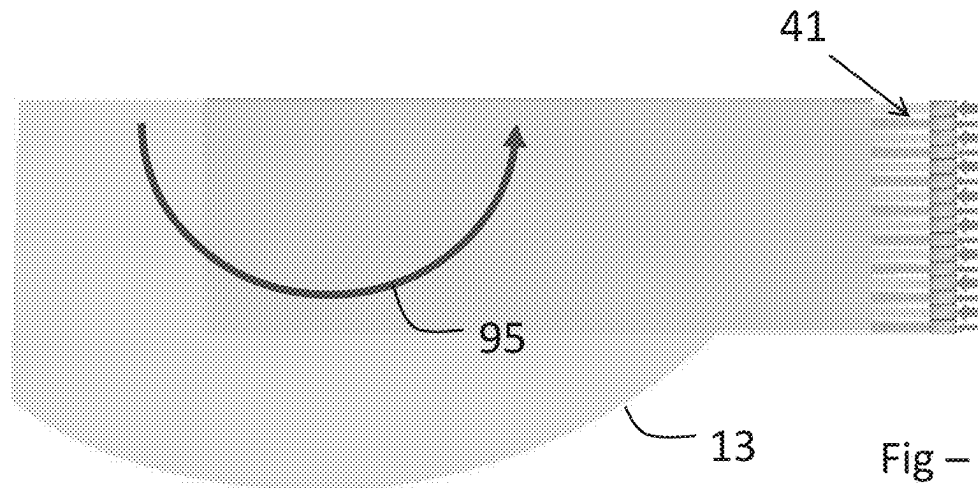


Fig. - 4b





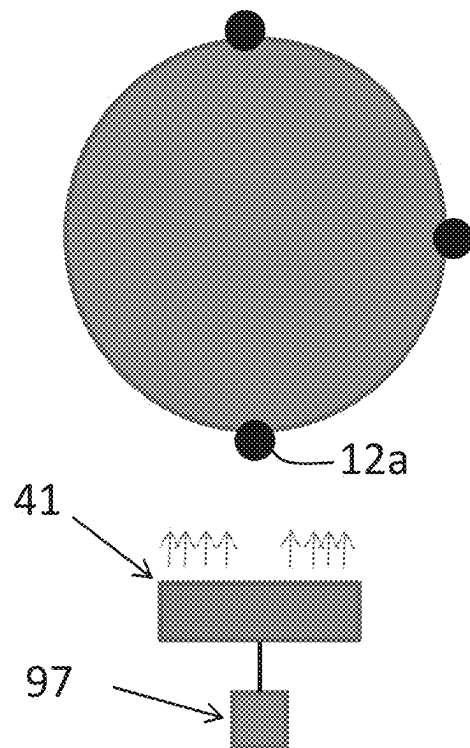


Fig – 5d

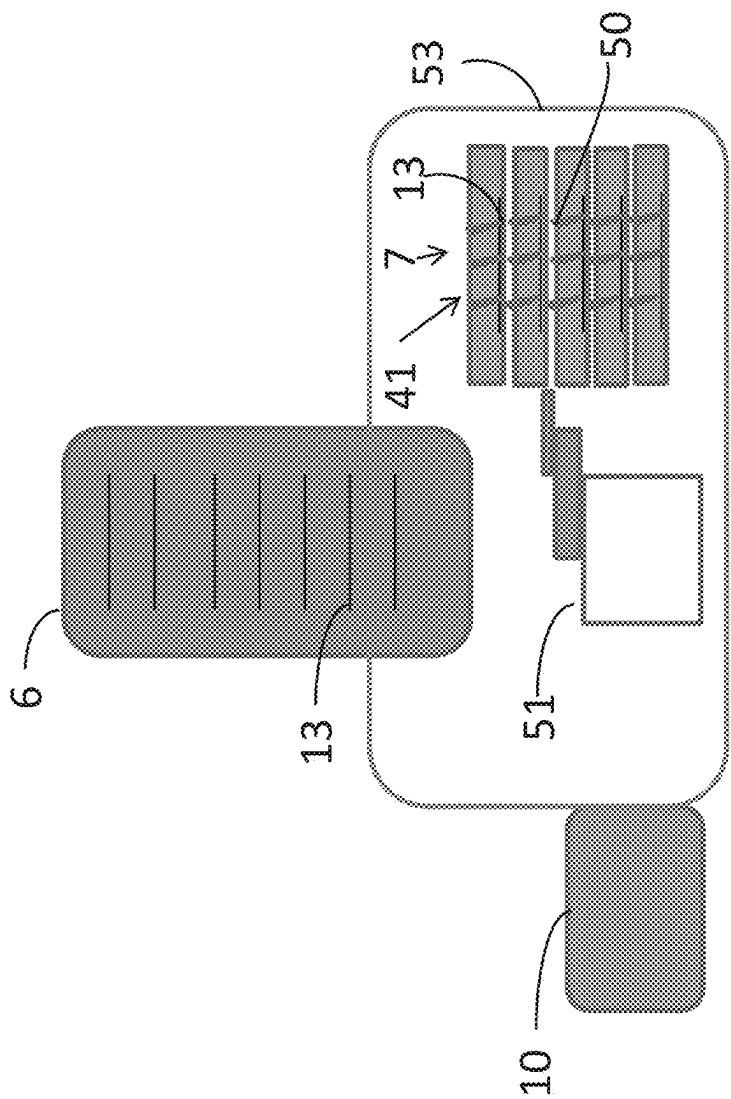


Fig--6

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**SUBSTRATE PROCESSING APPARATUS AND
METHOD FOR PROCESSING SUBSTRATES****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is continuation of, and claims priority to U.S. patent application Ser. No. 16/797,346 filed Feb. 21, 2020 titled SUBSTRATE PROCESSING APPARATUS AND METHOD FOR PROCESSING SUBSTRATES, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/809,191, filed Feb. 22, 2019, and titled SUBSTRATE PROCESSING APPARATUS AND METHOD FOR PROCESSING SUBSTRATES, the disclosures of which are hereby incorporated by reference in their entirety.

FIELD

The present disclosure generally relates to a substrate processing apparatus for processing a plurality of substrates. More particularly, the disclosure relates to a substrate processing apparatus, comprising:

a first reactor constructed and arranged to process a rack with a plurality of substrates therein;

a second reactor constructed and arranged to process a substrate; and,

a substrate transfer device constructed and arranged to transfer substrates to and from the first and second reactor.

BACKGROUND

Substrate processing apparatus also called furnaces may be provided with reaction chambers to create fine dimension structures, such as integrated circuits, on a plurality of substrates supported in a rack. In a typical substrate treatment step the substrates in the rack may be heated. Further, reactant gases may be passed over the heated substrate, causing the deposition of a thin layer of the reactant material on the substrate to be treated.

A series of treatment steps to deposit a layer on a substrate is called a recipe. Through subsequent deposition, doping, lithography, etch and other processes the layers are made into integrated circuits, producing from tens to thousands or even millions of integrated devices, depending on the substrate size and the circuits' complexity.

Various process parameters are carefully controlled to ensure the high quality of the resulting deposited layers. One such critical parameter is the substrate temperature during each recipe step. During chemical vapor deposition (CVD), for example, the deposition gases react within particular temperature windows and deposit on the substrate. Different temperatures result in different deposition rates and quality and therefor it is important to accurately control the substrate temperature to bring the substrate to the desired temperature before the reaction treatment begins.

The substrate, however, may comprise features that are temperature sensitive and therefor the temperature may be limited to a certain maximum to avoid damage to those sensitive features. This may lead to contradicting requirements in which for productivity, quality and/or reactivity the temperature should be high, while to avoid damage to the features on the substrate, the temperature should remain low.

By irradiating the top surface of the substrates with ultraviolet radiation, it may be possible to provide energy to

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the top surface for certain processes while not overheating the substrate. The energy may lead to a better quality of the deposited layer.

Incorporating an illumination system constructed and arranged to irradiate ultraviolet radiation into a furnace which is also used for a deposition process may be difficult, since the deposition process may also deposit on parts of the illumination system deteriorating the transmission of ultraviolet radiation.

SUMMARY OF THE INVENTION

This summary is provided to introduce a selection of concepts in a simplified form. These concepts are described in further detail in the detailed description of example embodiments of the disclosure below. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

According to an objective, it may be desirable to provide a substrate processing apparatus, comprising a first reactor constructed and arranged to process a rack with a plurality of substrates therein; a second reactor constructed and arranged to process a substrate; and, a substrate transfer device constructed and arranged to transfer substrates to and from the first and second reactor. The second reactor may be provided with an illumination system constructed and arranged to irradiate ultraviolet radiation within a range from 100 to 500 nanometers onto a top surface of at least a substrate in the second reactor.

By irradiating the surface of the substrates with ultraviolet radiation in the second reactor, it may be possible to provide energy on the top surface. This energy may be provided, while the risk of overheating the substrate is minimized. The energy may improve the quality of the deposited layer.

The illumination system may be constructed and arranged to irradiate ultraviolet radiation with a range from 100 to 500, preferably 150 to 400, and even more preferably 170 to 300 nanometers. The first reactor may comprise an inlet constructed and arranged to provide a first precursor in the first reactor to deposit a layer on the substrates in the rack.

According to an embodiment, there may be provided a method of processing a substrate comprising:

providing a substrate in a rack with a plurality of substrates;

loading the rack with a plurality of substrates in a first reactor;

providing a first precursor in the first reactor to deposit a layer on the substrates;

unloading the rack with a plurality of substrates from the first reactor;

transferring a substrate with the deposited layer to the second reactor; and,

illuminating the deposited layer of the substrate in the second reactor with ultraviolet radiation within a range from 100 to 500 nanometers.

For purposes of summarizing the invention and the advantages achieved over the prior art, certain objects and advantages of the invention have been described herein above. Of course, it is to be understood that not necessarily all such objects or advantages may be achieved in accordance with any particular embodiment of the invention. Thus, for example, those skilled in the art will recognize that the invention may be embodied or carried out in a manner that achieves or optimizes one advantage or group of advantages

as taught or suggested herein without necessarily achieving other objects or advantages as may be taught or suggested herein.

All of these embodiments are intended to be within the scope of the invention herein disclosed. These and other embodiments will become readily apparent to those skilled in the art from the following detailed description of certain embodiments having reference to the attached figures, the invention not being limited to any particular embodiment(s) disclosed.

BRIEF DESCRIPTION OF THE FIGURES

It will be appreciated that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help improve understanding of illustrated embodiments of the present disclosure.

FIG. 1 shows, diagrammatically and partially exposed, a perspective view of an apparatus suitable for the illumination system according to an embodiment;

FIG. 2 shows, diagrammatically, a plan view of the apparatus according to FIG. 1;

FIG. 3 shows, diagrammatically, a plane view of a cross-section of a substrate rack with a substrate that is illuminated with an illumination system according to an embodiment;

FIG. 4a shows an illumination system formed in a helical form according to an embodiment;

FIG. 4b depicts a portion of a gas discharge lamp for use in an illumination system according to an embodiment;

FIGS. 5a-5d depict an illumination system transmitting radiation according to a further embodiment;

FIG. 6 depicts schematically a side view of a cross section of a substrate processing apparatus according to a further embodiment.

DETAILED DESCRIPTION OF THE FIGURES

Although certain embodiments and examples are disclosed below, it will be understood by those in the art that the invention extends beyond the specifically disclosed embodiments and/or uses of the invention and obvious modifications and equivalents thereof. Thus, it is intended that the scope of the invention disclosed should not be limited by the particular disclosed embodiments described below.

An apparatus 1 suitable for the illumination system according to an embodiment may be indicated in FIGS. 1 and 2. Said apparatus 1 may comprise a housing 2 and may in general be installed partially or completely in a so-called "clean room." In addition to housing 2, partitions 3, 4 and 5 may be present, as can be seen in particular from FIG. 2. Housing 2 may delimit, with partition 3, reactor area 21. A substrate handling chamber 22 may be delimited between housing 2 and partitions 3, 4. A cassette handling chamber 23 may be delimited between partitions 4 and 5 and housing 2. The apparatus 1 may further comprise a cassette introduction portion 33.

First and second reactor chambers 6, 7, may be arranged in reactor area 21. Said reactor chambers may be positioned vertically and substrate racks, indicated by 12, filled with substrates 13, may be moved into the reactor chambers 6, 7 in the vertical direction from below. To this end each reactor chamber may have a rack handler comprising an insertion arm 14, which is movable in the vertical direction with the aid of a spindle 38. Only one insertion arm 14 can be seen

in the drawing of FIG. 1; however, there may be two insertion arms 14 on both sides of the apparatus.

The substrate rack 12 may be provided at the bottom with an insulating plug, which is not indicated in more detail, which provides a seal with respect to the reactor chamber. The reactor chamber may be referred to as a furnace and may be provided with a heater to heat the substrates.

The rack handler may comprise a rotary platform 11, provided with cut-outs 15, arranged in the reaction area 21. Said cut-outs 15 may be shaped such that, if the cut-outs 15 have been brought into the correct position, arm 14 is able to move up and down through the cut-outs. On the other hand, the diameter of the bottom of the substrate rack may be such that said diameter is larger than the cut-out 15 in the platform 11, so that when the arm 14 moves downwards from the position shown in FIG. 1 the substrate rack 12 may be placed on rotary platform 11 and may be removed therefrom again in a reverse operation.

The substrate racks 12 may be fed to both reactor chambers 6 and 7 with the rack handler. It may be possible to perform a successive treatment so that one rack is first treated in the first reactor and secondly in the second reactor. It may also be possible to allow parallel groups of substrate racks 12 to be treated exclusively by reactor chamber 6 and exclusively by reactor chamber 7. Said substrate racks 12 may be provided with substrates 13.

Substrates 13 may be supplied in (transport) cassettes 10 which, from the introduction portion 33, may be placed in store 8 through a closable opening 34 with the aid of arm 31 of the cassette handling robot 35. Arm 31 may be provided with a bearing surface 32 which has dimensions a little smaller than those of the series of cut-outs 26 in rotary platforms 27. A number of such rotary platforms may be provided one above the other in the vertical direction in store 8. Arm 31 may be movable in the vertical direction with the aid of cassette handling robot 35. Arm 31 may be mounted such that said arm is able not only to pick up or remove cassettes from or to introduction portion 33 to or from store 8, but also to make it possible to move cassettes from or to store 8 to or from rotary platform 30.

Said rotary platform 30 may be constructed such that on rotation the cassette is placed against partition 4 where an opening 37 has been made so that, after opening the cassettes, substrates can be taken one by one from the cassette concerned with the aid of arm 24 of a substrate handler and can be placed in the substrate rack 12 located in substrate handling chamber 22. Said substrate rack 12 is supported by a hinged arm 16 being part of the rack handler and provided with a bearing surface 17 at the end, the dimensions of which are somewhat smaller than those of cut-outs 15 of rotary platform 11. Said arm 16 may be able to move with the substrate rack through a closable opening in partition 3 by rotation about rotation point 18. A closure may be provided in order to be able to close opening 19 between reaction area 21 and substrate handling chamber 22.

An operator or an automated cassette transport system (not shown), may load store 8 by introducing a number of cassettes on introduction portion 33. Control operations may be done on panel 36. Cassettes 10 may be transferred from the introduction portion 33 with the aid of arm 31 into the storage compartments 9 made for these cassettes in store 8. By starting from the lowest position for removing the relevant cassette 10 from portion 33 through the opening 34, said cassette can be moved upwards for moving into a higher compartment 9 of the store 8 by the cassette handling robot 35. By rotation of the store 8, it is possible to fill various compartments 9 with cassettes 10.

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The cassettes 10 concerned may be removed from the store by arm 31 and placed on rotary platform 30. The cassettes are rotated on the rotary platform 30 and placed with their door against partition 4. The door of the cassette may be removed with a door opener. With the aid of arm 24, the substrates may be removed substrate by substrate and placed in substrate rack 12 placed on swing arm 16 with the substrate handler.

In the interim the rotary platform 11 may be able to move in the reactor area 21 in an optimum manner with regard to the treatments to be carried out on the substrates present inside the reactor area 21. After substrate rack 12 has been filled in the substrate handling chamber 22 and may become available to one of the reactor chambers 6, 7, opening 19, which was closed up to this time, is opened and said freshly filled substrate rack 12 may be placed on rotary platform 11. Said rotary platform 11 may then move one position and the filled substrate rack 12 may be removed from platform 11 with the help of insertion arm 14 into the reactor chambers 6, 7. Treated substrates in a finished rack may be lowered on said filled platform 11. Said substrates execute a movement counter to the above to end up in the cassettes.

The substrate rack 12 with the fresh substrate may be fed to reactor chamber 6 or 7 with the insertion arms 14 and may be treated in said chamber. The treatment may comprise an increase of the temperature of the substrates in the substrate rack 12 with a heater. It is important to accurately control the substrate temperature to bring the substrate to the desired temperature before the treatment begins to get the right productivity.

The substrate may comprise features that are temperature sensitive and therefor the temperature may be limited to a certain maximum to avoid damage to those sensitive features. This may lead to contradicting requirements in which for reactivity the temperature of the substrate may desirably be high, while the temperature of the substrate may desirably be low to avoid damaging the temperature sensitive features on the substrate.

It may be possible to perform a successive treatment in the reactor chamber 6, 7. The substrate rack 12 with substrates processed in the first reactor 6 may be transferred to the second reactor 7 for further treatment. The second reactor 7 may for example comprise an illumination system constructed and arranged to irradiate ultraviolet radiation within a range from 100 to 500 nanometers onto a top surface of at least one of the substrates in the substrate rack from a side of the substrate rack. The illumination system may be constructed and arranged to radiate ultraviolet radiation with a range from 100 to 500, preferably 150 to 400, and even more preferably 170 to 300 nanometers. By irradiating the top surface of the substrates from the side with ultraviolet radiation it may be possible to provide energy to the top surface for certain processes.

The energy may increase the reactivity on the top surface. This increase of reactivity may be accomplished while not overheating the substrate so that temperature sensitive features on the substrate may not get damaged. The increase of reactivity may lead to a better quality of the deposited layer and/or a higher productivity of the apparatus. It may also lead to certain processes becoming possible at a temperature on which before they were not possible because the reactivity was zero.

For example, the substrate processing apparatus may comprise a first reactor 6 constructed and arranged to process a rack with a plurality of substrates therein and a second reactor 7 may be constructed and arranged to process a substrate. The apparatus may have a substrate transfer

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device 51 comprising a rack and a substrate handler to transfer substrates to and from the first reactor 6. The substrate handler may be used to transfer substrates to the substrate holder of the second reactor 7 as well. The second reactor 7 may be provided with an illumination system 41 constructed and arranged to irradiate ultraviolet radiation within a range from 100 to 500 nanometers onto a top surface of at least a substrate in the second reactor 7.

The first reactor 6 may comprise an inlet constructed and arranged to provide a first precursor in the first reactor 6 to deposit a layer on the substrates in the rack 12. The first precursor may comprise silicon to deposit a silicon comprising layer on the substrates in the first reactor. For example, the first precursor may comprise silicohalides, metalorganicsilicon, trisilane, disilane or silane. The first precursor may comprise a metal selected from the group of aluminum, titanium and hafnium, zirconium to deposit a metal comprising layer on the substrates in the first reactor. For example, the first precursor may be TiCl_4 or TMA.

The first reactor 6 may comprise an inlet constructed and arranged to provide a second precursor in the first reactor 6 to react with the first precursor into the layer on the substrates in the rack 12 before the substrate is transferred to the second reactor for illumination. The second precursor may comprise nitrogen to deposit a nitrogen comprising layer on the substrates in the first reactor. For example, the second reactor may comprise NH_3 , N_2H_4 . The second precursor may comprise oxygen to deposit an oxide layer on the substrates in the first reactor. For example, the second precursor can include H_2O , O_3 , N_2O , and/or H_2O_2 .

The first and second precursor may deposit on the substrates in the first reactor with an atomic layer deposition process or with a chemical vapor deposition process.

After the layer is deposited, the rack may be moved down with the substrate transfer device and transferred from the first reactor 6 to the second reactor 7. The substrate processing apparatus may comprise a rack conveyer constructed and arranged to horizontally transfer a rack with substrates from the first reactor 6 to the second reactor 7. The rack may be moved up into the second reactor 7 with an elevator.

The second reactor 7 may be constructed and arranged to receive the substrate rack in the reaction chamber. The second reactor may have an illumination system constructed and arranged to irradiate ultraviolet radiation within a range from 100 to 500 nanometers onto a top surface of at least one of the substrates in the substrate rack. The illumination system may irradiate the substrates from a side of the substrate rack. The quality of the deposited layer may be increased by the ultraviolet radiation while not overheating the substrate. After treatment with ultraviolet radiation, the substrate may be moved back for a further layer to be deposited and the cycle of deposition and illumination may be repeated. If the layer is finished the substrates may be transferred out of the apparatus.

By having the deposition process in the first reactor 6 and the illumination system in the second reactor 7, the deposition process may not contaminate the illumination system in the second reactor 7. The transmission of ultraviolet radiation in the second reactor 7 may therefore be substantially unaltered during the lifetime of the apparatus.

FIG. 3 shows a cross-section of a substrate rack 12 with a substrate 13 that is illuminated from four sides in the second reactor 7 of FIGS. 1 and 2 provided with the illumination system 41. The illumination system 41 may comprise four parts, for example tubes 43, to irradiate ultraviolet radiation to the substrate 13 from multiple sides.

The illumination system **41** may be configured to irradiate ultraviolet radiation within a range from 100 to 500 nanometers.

The tubes **43** of the illumination system **41** may be elongated and extend in a direction perpendicular to the substrate surface. The tubes **43** of the illumination system **41** may extend over a part of the rack **12**, over the full length of the rack **12** or even a bit further. The tubes **43** of the illumination system may have a length of between 50 and 200 cm, preferably 75 and 150 cm to illuminate the substrates over the full length of the rack **12**.

The illumination system **41** for illuminating the substrate surface may have a power of between 5 W and 100 kW, preferably 300 W and 20 kW and even more preferably between 1 and 10 kW. The illumination system may have an efficiency of between 50 and 90% in the conversion of electrical energy to ultraviolet radiation. The illumination system may have a power output of between 0.05 W and 1 kW per centimeter, preferably between 3 and 200 W per centimeter and most preferably between 10 and 100 W per centimeter in the direction perpendicular to the substrates.

The substrate surface may receive a power between 0.1 and 200 milliwatt (mW)/cm², preferably between 1 and 100 mW/cm² and even more preferably between 5 and 80 mW. The illumination system may be constructed and arranged to radiate ultraviolet radiation with a range from 100 to 500, preferably 150 to 400, and even more preferably 170 to 300 nanometers. The rack **12** may have a length between 50 and 200 cm. The illumination system may comprise an optical wave guide to guide the radiation to the substrates. The optical wave guide may comprise an optical fiber. The illumination system may be provided with radiation reflecting surfaces to direct the ultraviolet radiation to the substrates.

The substrate **13** may be positioned in the substrate rack **12** which may comprise three struts comprising a plurality of spaced apart substrate holding provisions configured to hold the plurality of substrates in a spaced apart relationship. The rack **12** may have a maximum of between 50 and 200, preferably between 100 and 180 spaced apart substrate holding provisions along the struts for holding an equal amount of substrates.

For an optimal production, the rack may be filled until the maximum; however, to increase the power received on the substrates and to improve the uniformity of the radiation received over the surface of the substrate, the number of substrates in the rack **12** may be made lower than the maximum. For example, the rack may be provided with 10 to 80 of substrates in a spaced apart relationship. The distance between the substrates in the rack may in such case be between 5 to 200, preferably 20 to 140 and most preferable between 40 mm to 100 mm.

The struts may be elongated and extend in a direction perpendicular to the substrate surface. The plurality of substrates may be positioned parallel with each other in the substrate rack **12**. The configuration of the substrate rack **12** and the illuminations system **41** causes the illumination **41** to irradiate the ultraviolet radiation onto a top surface of at least one of the substrates in the substrate rack from a side of the substrate rack **12**. As depicted, the illumination system may comprise four parts to irradiate ultraviolet radiation from four sides to the substrate. Illuminating from four sides may improve the uniformity of the illumination received on the substrate. The illumination system may also have one, two, three or four parts, which illuminate the substrate surface.

Ultraviolet radiation may be creating a plasma in gasses through which it may traverse. The plasma may be helpful or unwanted for the processes running in the reaction chamber.

If the plasma is unwanted, the apparatus may be constructed and arranged to suppress plasma in the second reaction chamber **7**. The apparatus may also be constructed and arranged to obstruct the plasma from propagating into the second reaction chamber **7**. For example, by providing the apparatus with plasma shielding, e.g., conductive wiring or coatings, the plasma may be suppressed or obstructed before it reaches the reaction chamber. The apparatus may also be provided with a program, which when run on the apparatus, selects the gas, pressure range and/or power range such that the creation of a plasma in the interior of the second reaction chamber **7** may be suppressed.

FIG. **4a** shows an illumination system **41** to be provided to the second reaction chamber **7** formed in a helical form which may be used to illuminate the top surface of the substrates. The helically formed illumination system may be configured surrounding a substrate rack **12** with substrates **13**. The illumination system **41** may be a gas discharge lamp.

FIG. **4b** depicts a portion of a gas discharge lamp. Gas discharge lamps generate radiation by having an electric discharge between two electrodes through an ionized gas, e.g., a plasma in a tube **43**. Such lamps may use a noble gas such as argon, neon, krypton, and xenon or a mixture thereof and additionally even may use mercury, sodium, and metal halides in the mixture in the tube **43**. The electrons may be forced to leave atoms of the gas near an anode by the electric field applied between the two electrodes from which only one **45** is depicted, leaving these atoms positively ionized. Free electrons flow to the anode, while the cations flow to the cathode. The ions may collide with neutral gas atoms, which transfer their electrons to the ions. The atoms, having lost an electron during the collisions, ionize and speed toward the cathode while the ions, having gained an electron during the collisions, return to a lower energy state while releasing energy in the form of radiation, which is emitted in the direction of the substrate top surface of the substrate to transfer its energy into the top surface. The electrode **45** is mounted in a base **47** connected to the tube **43** and being provided with pins **49**.

FIGS. **5a** to **5d** depict an illumination system according to a further embodiment. FIG. **5a** depicts a side view on the illumination system **41** comprising individually controllable radiation sources, for example light emitting diodes, to control the power output for illuminating the substrates **13** from the side individually along a stack of substrates in a vertical direction. The illumination system **41** for emitting radiation beams in the direction of the substrates **13** may be positioned on a side of the rack **12**. The illumination system **41** may irradiate radiation beams from the side downward towards the top surface of the substrate **13**. As shown here, the illumination system only illuminates the top part of the rack **12**; however, in some cases, the illumination system **41** may be extended over the full length of the rack **12**.

The angle of the radiation beams may be between 60 to 90° preferably between 80 to 89.5° and even more preferably between 85 and 89° with respect to a line perpendicular to the top surface of the substrate **13**. The radiation beam of the illumination system **41** may be slightly parallel. The direction of the radiation beam of the illumination system may therefore be defined as the average direction of the radiation emitted by the illumination system **41**.

The apparatus may comprise reflectors (not shown) on the other side of the substrate rack with respect to the illumina-

nation system **41** to reflect radiation reflected of the substrates **13** back to the substrate surface. The reflectors may be retroreflectors to reflect the radiation beam back in the same direction as from which the radiation beam came. The reflector may comprise a material selected from the group of material comprising glass, steel, aluminum or polytetrafluoroethylene (PTFE) to direct the radiation to the substrates.

The reflector may be provided with a polarizer to change the polarization of the reflected light by 90 degrees to improve the absorption of the reflected light. The polarizer may be a thin plate with a thickness of $\frac{1}{8}$ th of the wavelength positioned in front of the reflector.

The illumination system may have a first and second group of individually controllable radiation sources **91**, **93**. The first group of individually controllable radiation sources **91** may be directed to a surface of the substrate **13** further away from the edge and have an increased power output with respect to the second group of individually controllable radiation sources **93** directed to top surface near the edge of the substrate **13**. The uniformity of the radiation intensity over the substrate surface may be increased in this way. If the radiation intensity is uniform over the substrate surface, the reactivity increase by the illumination system **41** over the substrate surface becomes the same, which is advantageously for process control.

As depicted, the illumination system **41** may be directly illuminating the substrate **13**; however, the reaction chamber may also be limited by a process tube in between the illumination system and the substrates **13**. The process tube may be forming a barrier for processing gasses and at least partially functioning as the radiation transmitting surface. The illumination system **41** may be provided outside the reaction chamber and may be constructed and arranged to irradiate the ultraviolet radiation through the radiation transmitting surface into the reaction chamber. The process tube may be protecting the illumination system **41** from the alleviated temperature and deposition products provided in the reaction chamber.

FIG. **5b** depicts a top view on the illumination system of FIG. **5a**. If the illumination system is provided from one side only, a portion of the substrate **13** may be directly illuminated. By providing the apparatus with a rotation device to rotate the substrate the substrate in a direction as depicted by the arrow **95**, it may be assured that the substrate **13** is uniformly illuminated.

The substrate rack **12** may be provided at the bottom with an insulating plug, which provides a seal with respect to the reaction chamber **6**, **7**, when the rack **12** is moved upward in the reaction chamber **6**, **7** (see FIGS. **1** and **2**). To increase the uniformity of the illumination by the illumination system **41** the insulating plug may be provided with a (rack) rotation device for rotating the rack **12** with substrates **13** around a vertical axis.

Rack rotation devices may be known from U.S. Pat. No. 9,018,567 B2, which is hereby incorporated herein by reference. The uniformity of the radiation intensity over the substrate surface may be increased in this way. If the radiation intensity is uniform over the substrate surface, the reactivity increase by the illumination system **41** over the substrate surface becomes the same, which is advantageous for process control.

FIG. **5c** depicts a problem that may arise in the apparatus using the illumination system **41** to illuminate the substrates **13** and having a (rack) rotation device for rotating the rack **12** with substrates **13** around a vertical axis of FIG. **5b**. The radiation of the illumination system **41** may illuminate and or heat-up a part **12a** of the substrate rack excessively. The

radiation may scatter from the substrate rack **12** through the ambient of the reaction chamber **6** illuminating parts of the apparatus that are not intended to be illuminated.

FIG. **5d** depicts an illumination system according to a further embodiment solving the problem of excessive illuminating and or heating-up a part **12a** of the substrate rack in FIG. **5c**. The substrate rack **12** may be rotated to achieve a more uniform illumination distribution and circumvent excessive heating.

Further, the information of the shape of the substrate rack **12** and the rotation position of the rack **12** that is available from the control system may be used to switch off or limit the power of the part of the illumination system **41** that will be hitting the aforementioned part **12a** of the rack **12**. A reduced amount of radiation may therefore be received by the part **12a** of the substrate rack **12** and less radiation may scatter from the substrate rack **12** through the ambient of the support member illuminating and heating up parts of the apparatus that are not intended to be illuminated or heated up.

The apparatus may comprise a power controller **97** to control the power of the illumination system **41** and the power controller may be programmed to adjust a radiation output of the illumination system **41** along the width of the substrate rack to avoid excess heating of the substrate rack.

FIG. **6** depicts schematically a side view of a cross section of a substrate processing apparatus with a first reactor **6** which is similar to the first reactor **6** of FIG. **1** and a second reactor **7**. The second reactor **7** may be provided with a substrate holder **50** constructed to hold a single substrate **13** and an illumination system **41** constructed and arranged above the holder to illuminate a substrate **13** in the holder **50** on the top side. The illumination system may irradiate ultraviolet radiation within a range from 100 to 500 nanometers onto a top surface of at least a substrate in the second reactor. The second reactor **7** may be provided with multiple, for example, five substrate holders **50** each constructed and arranged to hold a single substrate **13** under the illumination system **41**.

The apparatus may have a substrate transfer device **51** comprising a substrate handler to transfer substrates to and from the first and second reactor **6**, **7**. The substrate handler may be constructed and arranged to transfer substrates to and from spaced apart substrate holding provisions of a rack configured to hold a plurality of substrates in a spaced apart relationship. The first reactor **6** may be constructed and arranged to receive the substrate rack in the first reactor **6**. The substrate transfer device **51** may comprise an elevator constructed and arranged to move the rack in the first reactor **6**.

The substrate handler of the substrate transfer device **51** may also be used to transfer substrates to the substrate holder **50** of the second reactor **7**. In this case a single substrate is transferred to the substrate holder **50**. The illumination system **41** may comprise a radiation source (a light emitting diode, an excimer source (lamp or laser), a Mercury-vapor lamp, laser) irradiating ultraviolet radiation onto the top surface of the substrate **13**.

The substrate handler may be constructed and arranged to transfer substrates in a first direction towards the first reactor **6** and in a second direction towards the second reactor **7**. The first and second direction may form an angle of 90 to 180 degrees with each other. The substrate transfer device may be provided in a substrate transfer device chamber **53** which is provided with an inert space.

The substrate transfer device chamber may be provided with an inert space creation system, e.g., a nitrogen purge

system, a vacuum evacuation system or a low oxygen system to create an inert space in the substrate transfer device chamber 55. The inert space in the substrate transfer device chamber 53 may circumvent oxidation of layers deposited on substrates 13 in the first reactor 6 during transfer to the second reactor 7 for treatment by the illumination system 41.

The second reactor 7 may also be provided with an inert space creation system, e.g., a nitrogen purge system, a vacuum evacuation system or a low oxygen system to create an inert space in the second reactor for the same reason.

The second reactor 7 may be provided with a cleaning inlet to provide a cleaning gas (e.g., etch reactant) in the second reactor to clean outgassing products away in the second reactor or to prepare a substrate.

The substrate processing apparatus may be used to process a substrate by providing a substrate in a rack with a plurality of substrates and loading the rack with a plurality of substrates in the first reactor 6 with the substrate transfer device 51. Subsequently a first, and optionally also, a second precursor may be provided in the first reactor to deposit a layer on the substrates.

The rack with the processed substrates may be lowered from the first reactor 6 with the elevator of the substrate transfer device 51. The substrates with the deposited layer may be transferred from the rack to the substrate holders 50 of the second reactor 7 with the substrate handler of the substrate transfer device 51. The deposited layer of the substrate on the substrate holder 50 in the second reactor 7 may be illuminated with ultraviolet radiation within a range from 100 to 500 nanometers.

The illuminated substrates 13 may be transferred from the second reactor 7 to the first reactor 6 with the substrate transfer device 51 again to deposit a further layer on the substrates 13. After deposition the substrate with the deposited further layer may be transferred to the second reactor 7 again for illumination with ultraviolet radiation within a range from 100 to 500 nanometers. In this way thin layers of freshly deposited material may be treated with ultraviolet radiation repeatedly. The latter may be advantageous when the deposited material has a limited transmissivity for ultraviolet radiation making it difficult to improve the quality of the layer in depth. If the substrates are ready the substrates may be transferred to the cassette 10 with the substrate transfer device 51.

A first and second precursor may be provided in the first reactor 6 to react with each other to form the layer on the substrates. The layers may be deposited by atomic layer deposition (ALD) or chemical vapor deposition (CVD) reactions. The ultraviolet illumination system may be used to improve the quality of layers deposited by atomic layer deposition (ALD) or chemical vapor deposition (CVD).

Before the substrates are provided in the first reactor 6 to deposit a layer thereon, the substrates may be illuminated in the second reactor with ultraviolet radiation within a range from 100 to 500 nanometers to prepare the substrate. The second reactor may be provided with a cleaning gas (e.g., etch reactant) to prepare the substrate.

A complementary periodical in situ clean with etch gases may be a required in the second reactor to clean the radiation transmitting or reflecting surface in the apparatus. The apparatus may comprise an etching system. The etching system may comprise a fluid storage, a control system and a valve. The control system may be provided with a program when run on the control system to improve the transmissivity of the radiation transmitting or reflecting surface of the second reactor.

An etching fluid may be stored in the fluid storage of the etching system. The control system may be controlling a valve for providing the etching fluid in the reaction chamber 6. The control system may control the valve to provide the etching fluent, i.e., etchant in the reaction chamber, so as to etch a layer deposited on the radiation transmitting or reflecting surface away to improve the transmissivity of the surface.

The etching fluid may be chloride (Cl_2), boriumchloride (BCl_3), hydrogenchloride (HCl), tetrafluoromethane (CF_4), nitrogentrifluoride (NF_3), hydrogenbromide (HBr), sulfur hexafluoride (SF_6), fluoride (F_2), chlorine trifluoride (ClF_3) or an ashing component created by ultraviolet radiation in combination with an hydrogen or oxygen comprising gas, such as hydrogen or oxygen.

The particular implementations shown and described are illustrative of the invention and its best mode and are not intended to otherwise limit the scope of the aspects and implementations in any way. Indeed, for the sake of brevity, conventional manufacturing, connection, preparation, and other functional aspects of the system may not be described in detail. Furthermore, the connecting lines shown in the various figures are intended to represent exemplary functional relationships and/or physical couplings between the various elements. Many alternative or additional functional relationship or physical connections may be present in the practical system, and/or may be absent in some embodiments.

It is to be understood that the configurations and/or approaches described herein are exemplary in nature, and that these specific embodiments or examples are not to be considered in a limiting sense, because numerous variations are possible. The specific routines or methods disclosed herein may represent one or more of any number of processing strategies. Thus, the various acts illustrated may be performed in the sequence illustrated, in other sequences, or omitted in some cases.

The subject matter of the present disclosure includes all novel and nonobvious combinations and sub combinations of the various processes, systems, and configurations, and other features, functions, acts, and/or properties disclosed herein, as well as any and all equivalents thereof.

Although certain embodiments and examples are disclosed herein, it will be understood by those in the art that the invention extends beyond the specifically disclosed embodiments and/or uses of the invention and obvious modifications and equivalents thereof. Thus, it is intended that the scope of the invention disclosed should not be limited by the particular disclosed embodiments described herein. The illustrations presented herein are not meant to be actual views of any particular material, structure, or device, but are merely idealized representations that are used to describe embodiments of the disclosure.

As used herein, the term "substrate" or "wafer" may refer to any underlying material or materials that may be used, or upon which, a device, a circuit, or a film may be formed. The term "semiconductor device structure" may refer to any portion of a processed, or partially processed, semiconductor structure that is, includes, or defines at least a portion of an active or passive component of a semiconductor device to be formed on or in a semiconductor substrate. For example, semiconductor device structures may include, active and passive components of integrated circuits, such as, for example, transistors, memory elements, transducers, capacitors, resistors, conductive lines, conductive vias, and conductive contact pads.

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The invention claimed is:

1. A reaction chamber comprising:
a rack configured to hold a plurality of substrates, wherein the plurality of substrates are aligned vertically in the rack;
an illumination system configured to produce ultraviolet radiation within a range from 100 to 500 nanometers, wherein the illumination system is constructed and arranged to irradiate ultraviolet radiation onto a top surface of each of the plurality of substrates in the rack in the reaction chamber, wherein the illumination system is configured such that the average direction of the ultraviolet radiation is at an angle in the range of 80 to 89.5 degrees with respect to a line perpendicular to the top surfaces of the plurality of substrates; and
a power controller configured to control the power of the illumination system, wherein the power controller is programmed to adjust a radiation output of the illumination system along a width of the rack.
2. The reaction chamber of claim 1, wherein the illumination system comprises at least three elongated radiation sources spaced about a periphery of the rack when the rack is positioned within the reaction chamber, each of the elongated radiation sources extending in a direction perpendicular to the top surfaces of the plurality of the substrates and extending over a full length of the rack to illuminate the substrates over the full length of the rack.
3. The reaction chamber of claim 2, wherein each of the at least three elongated radiation sources is arranged to generate a radiation beam from the side of the rack downward to one of the top surfaces of the plurality of substrates, wherein the average direction of radiation of the radiation beam is at an angle in the range of 60 to 90 degrees with respect to a line perpendicular to the top surfaces of the plurality of substrates.
4. The reaction chamber of claim 3, wherein the average direction of radiation of the radiation beam is at an angle in the range of 80 to 89.5 degrees with respect to a line perpendicular to the top surfaces of the plurality of substrates.
5. The reaction chamber of claim 3, wherein the average direction of radiation of the radiation beam is at an angle in the range of 85 to 89 degrees with respect to a line perpendicular to the top surfaces of the plurality of substrates.
6. The reaction chamber of claim 2, wherein each of the elongated radiation sources comprises a tube configured and wherein the illumination system illuminates at least three different sides of the substrate.
7. The reaction chamber of claim 1, wherein the illumination system is configured to produce ultraviolet radiation within a range from 170 to 300 nanometers.
8. The reaction chamber of claim 2, wherein each of the elongated radiation sources comprises a radiation source irradiating ultraviolet radiation onto top surfaces of a plurality of the substrates.
9. The reaction chamber of claim 1, the illumination system is constructed and arranged to irradiate ultraviolet radiation within a range from 100 to 500 nanometers onto the top surface from a side of the rack.
10. The reaction chamber of claim 1, further comprising an optical wave guide.

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11. The reaction chamber of claim 1, wherein the reaction chamber is provided with an inert space creation system to create an inert space in the reaction chamber.

12. The reaction chamber of claim 1, wherein the reaction chamber is provided with a cleaning inlet to provide a cleaning gas in the reaction chamber to clean outgassing products away in the reaction chamber or prepare a substrate.

13. The reaction chamber of claim 1, wherein the illumination system comprises a plurality of individually controllable radiation sources arranged in at least one linear set along a line, wherein the line is perpendicular to the top surfaces of the plurality of substrates along a full length of the rack, wherein the power controller is programmed to adjust a radiation output of at least one of the plurality of individually controllable radiation sources along a width of the rack.

14. The reaction chamber of claim 13, wherein each of the individually controllable radiation sources comprises a light emitting diode.

15. The reaction chamber of claim 13, wherein each of the plurality of individually controllable radiation sources is arranged to generate a radiation beam from the side of the rack downward to one of the top surfaces of the plurality of substrates, wherein the average direction of radiation of the radiation beam is at an angle in the range of 60 to 90 degrees with respect to a line perpendicular to the top surfaces of the plurality of substrates.

16. The reaction chamber of claim 15, wherein the average direction of radiation of the radiation beam is at an angle in the range of 80 to 89.5 degrees with respect to a line perpendicular to the top surfaces of the plurality of substrates.

17. The reaction chamber of claim 15, wherein the average direction of radiation of the radiation beam is at an angle in the range of 85 to 89 degrees with respect to a line perpendicular to the top surfaces of the plurality of substrates.

18. The reaction chamber of claim 13, wherein the individually controllable radiation sources comprise a first group of the individually controllable radiation sources and a second group of the individually controllable radiation sources, wherein the first group of the individually controllable radiation sources is configured to direct radiation towards a first portion of the top surfaces, and wherein the second group of the individually controllable radiation sources is configured to direct radiation towards a second portion of the top surfaces.

19. The reaction chamber of claim 13, further comprising a rotation device to rotate the plurality of substrates in the reaction chamber about a vertical axis extending through the centers of the plurality of substrates, whereby each of the plurality of substrates is more uniformly illuminated by the illumination system.

20. The reaction chamber of claim 1, wherein the illumination system comprises a helical shaped radiation source and wherein the rack is positionable within a center of the helical shaped radiation source, whereby the ultraviolet radiation is directed inward from sides of the rack onto the top surface of each of the plurality of substrates.

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